

Geometric Distribution

Geometric distribution comes under the category of discrete random variable distributions. The geometric distribution gives the probability that the first occurrence of success requires k independent trials, each with success probability p . If the probability of success on each trial is p , then the probability that the k th trial is the first success is -

$$P(X = k) = (1 - p)^{k-1} p$$

The geometric distribution is an appropriate model if the following assumptions are true.

- The phenomenon being modeled is a sequence of independent trials.
- There are only two possible outcomes for each trial, often designated success or failure.
- The probability of success, p , is the same for every trial.

If these conditions are true, then the geometric random variable X is the count of the number of failures before the first success.

The cdf of the distribution can be defined as :

$$F(k) = P(X \leq k) = 1 - (1 - p)^k$$

$$P(X > k) = (1 - p)^k$$

Expected value : $E[X] = \frac{1}{p}$

Variance : $\text{Var}[X] = \frac{(1 - p)}{p^2}$

Questions:

1. Suppose that the no. of runs scored of a delivery is uniform in $\{1, 2, 3, 4, 5, 6\}$ independent of what happens in other deliveries. A batsman needs to bat till he hits a six. What is the probability that he needs fewer than 6 deliveries to do so ?

Solution -

We can consider success to be hitting a six.

$$P(\text{hitting a six}) = \frac{1}{6}$$

$$P(\text{other than six}) = \frac{5}{6}$$

$$\begin{aligned} \text{He needs fewer than 6 deliveries so } P(x \leq 5) &= 1 - \left(1 - \frac{5}{6}\right)^5 \\ &= 0.59 \end{aligned}$$

2. A light bulb manufacturing factory finds 3 in every 60 light bulbs defective. Calculate what will be the probability that the first defective light bulb will be found when the 6th one is tested?

Solution -

As the probability of the first defective light bulb needs to be determined hence, this is a geometric distribution.

$$P(\text{defective bulb}) = \frac{3}{60} = 0.05$$

$$P(X = 6) = \left(1 - 0.05\right)^{6-1} 0.05$$

$$P(X = 6) = 0.0386$$

Probability (first defective light bulb is found on the 6th trial) is 0.0368.

3. A Football match between Argentina and Portugal is going on and till the last minute none of the teams has scored a single goal, so the match referee decided to go for a penalty shoot. From Argentina Messi will go for penalty all the time and from Portugal, Ronaldo will go for penalty each time. There is 40% chance that Messi can convert penalty into goal whereas 70% chance that Ronaldo can convert penalty into goal. Each penalty shoot is independent of previous one. The two players will go for penalty shoot alternatively and Ronaldo will start the penalty. If Ronaldo hits the penalty first then Portugal will win and Messi won't be allowed to do the penalty.

1. What is the probability that Messi wins before the third round of penalty ?

Solution -

As the penalty shoot out is started by Ronaldo, Messi can win only in the 2nd round if he has to win before the 3rd round.

Now,

$$P(\text{Messi scoring}) = 0.4 \quad P(\text{Messi not scoring}) = 0.6$$

$$P(\text{Ronaldo scoring}) = 0.7 \quad P(\text{Ronaldo not scoring}) = 0.3$$

$$\begin{aligned} P(\text{Messi wins before round 3}) &= \text{Ronaldo failing 1st penalty} * \text{Messi scoring 2nd one} \\ &= 0.3 * 0.4 \\ &= 0.12 \end{aligned}$$

2. What is the probability that Ronaldo will win ?

Solution -

$P(\text{Ronaldo win}) = (\text{Ronaldo score}) \text{ or } (\text{Ronaldo not score} * \text{Messi not score} * \text{Ronaldo not score}) \text{ or } (\text{Ronaldo not score} * \text{Messi not score} * \text{Ronaldo not score} * \text{Messi not Score} * \text{Ronaldo score}) \text{ and so on.....}$

This would generate an infinite series.....

$$\begin{aligned} P(\text{Ronaldo win}) &= 0.7 + (0.3 \times 0.6 \times 0.7) + (0.3 \times 0.6 \times 0.3 \times 0.6 \times 0.7) \dots \\ &= 0.7 + 0.18 \times 0.7 + 0.18^2 \times 0.7 \dots \\ &= \frac{a}{1-r} \\ &= \frac{0.7}{1-0.18} \\ &= 0.85 \end{aligned}$$